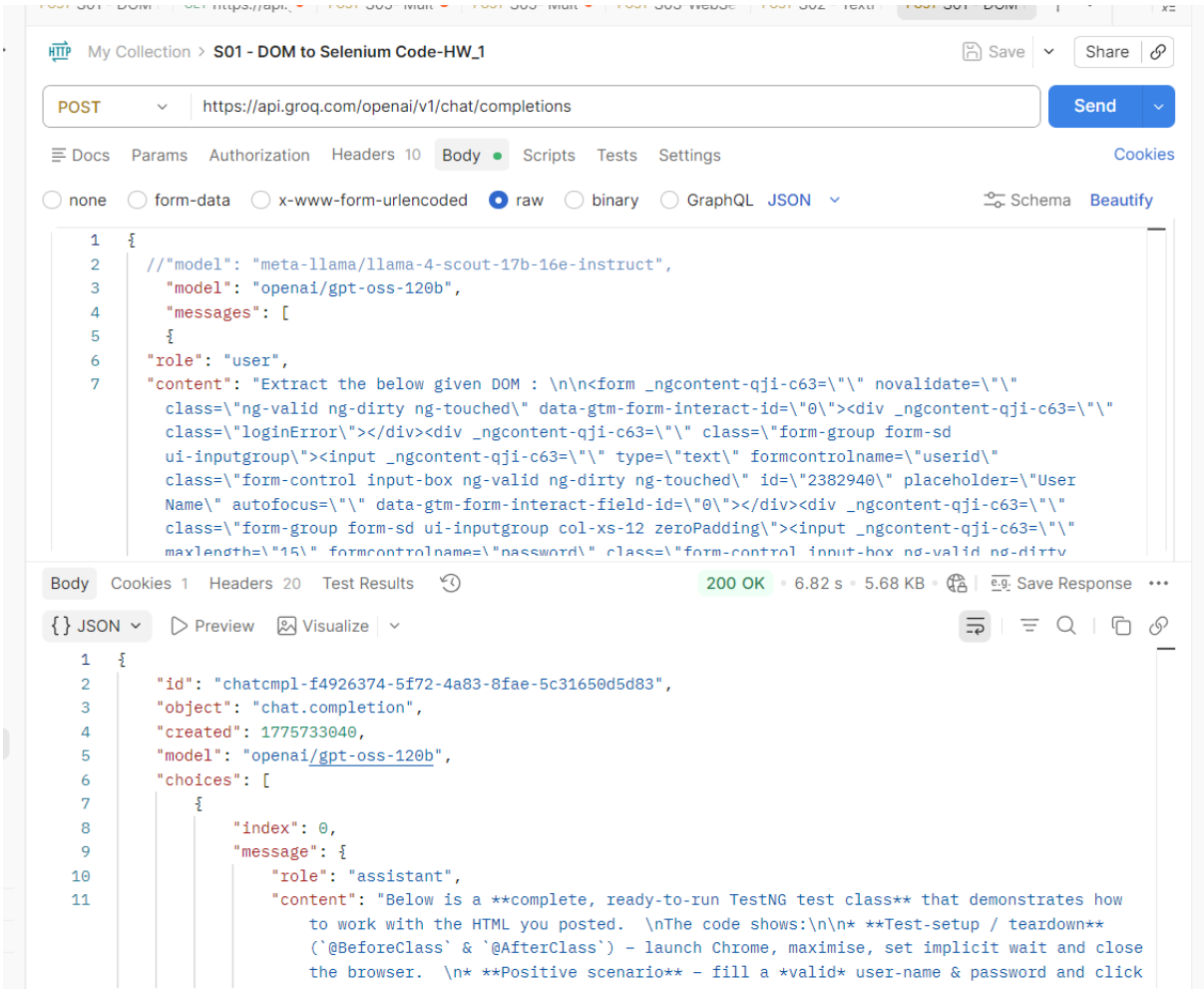
# 1. S01 - DOM to Selenium Code-HW

Note : Used IRCTC login page DOM




## (a) Request


```
{
  //"model": "meta-llama/llama-4-scout-17b-16e-instruct",
    "model": "openai/gpt-oss-120b",
    "messages": [
    {
  "role": "user",
  "content": "Extract the below given DOM : \n\n<form _ngcontent-qji-c63=\"\" novalidate=\"\" class=\"ng-valid ng-dirty ng-touched\" data-gtm-form-interact-id=\"0\"><div _ngcontent-qji-c63=\"\" class=\"loginError\"></div><div _ngcontent-qji-c63=\"\" class=\"form-group form-sd ui-inputgroup\"><input _ngcontent-qji-c63=\"\" type=\"text\" formcontrolname=\"userid\" class=\"form-control input-box
```

ng-valid ng-dirty ng-touched\" id=\"2382940\" placeholder=\"User Name\" autofocus=\"\" data-gtm-form-interact-field-id=\"0\"> <div _ngcontent-qji-c63=\"\" class=\"form-group form-sd ui-inputgroup col-xs-12 zeroPadding\"><input _ngcontent-qji-c63=\"\" maxlength=\"15\" formcontrolname=\"password\" class=\"form-control input-box ng-valid ng-dirty ng-touched\" id=\"2931697\" placeholder=\"Password\" type=\"password\" data-gtm-form-interact-field-id=\"1\"><i _ngcontent-qji-c63=\"\" class=\"password-icon fa fa-eye\"></i> <div _ngcontent-qji-c63=\"\" class=\"form-group form-sd\"><a _ngcontent-qji-c63=\"\" href=\"#\" onclick=\"return false;\" class=\"ui-inputgroup-addon\" style=\"color: #1457a7; padding-left: 0px; padding-right: 0px;\"><label _ngcontent-qji-c63=\"\" style=\"font-size: 11px;\"><strong _ngcontent-qji-c63=\"\">FORGOT ACCOUNT DETAILS?</label></a> <!----><div _ngcontent-qji-c63=\"\" class=\"form-group form-sd\" style=\"margin-bottom: 13px;\"><div _ngcontent-qji-c63=\"\"><input _ngcontent-qji-c63=\"\" type=\"checkbox\" name=\"otpLogin\" id=\"otpLogin\" formcontrolname=\"otpLogin\" aria-label=\"Booking using OTP\" class=\"css-checkbox ng-untouched ng-pristine ng-valid\"><label _ngcontent-qji-c63=\"\" for=\"otpLogin\" class=\"css-label_c t_c\" style=\"color: #082b71; font-size: small;\">Visually impaired users may select this option to receive OTP instead of CAPTCHA</label> <span _ngcontent-qji-c63=\"\"><button _ngcontent-qji-c63=\"\" type=\"submit\" class=\"search_btn train_Search train_Search_custom_hover\" style=\"padding: 10px 14px;\">SIGN IN</button></form>\n\n\nand provide me the selenium java code. \n\nInstructions: \n- Use TestNG Framework\n- Provide meaningful name for the method\n- Include Positive and Negative scenarios\n- Provide clear and meaningful comments for each Method"
}
  ],
  "temperature": 0.1,
  "include_reasoning":true
}

**Response:**

{
  "id": "chatcmpl-e9915bcc-fc3d-4519-92cb-b8f71f0bbdf7",
  "object": "chat.completion",
  "created": 1775718158,
  "model": "openai/gpt-oss-120b",
  "choices": [
    {
      "index": 0,

"message": {

"role": "assistant",

"content": "Below is a **complete, ready-to-run TestNG test class** that works with the HTML fragment you posted. \nThe class demonstrates:\n\n* **Positive scenario** – a successful sign-in with a valid user-name / password. \n* **Negative scenarios** – \n * empty user-name, \n * empty password, \n * wrong credentials, \n * trying to submit the form while the **“Visually impaired users … OTP”** checkbox is ticked (just to show the element can be toggled). \n\nAll elements are located with **stable locators** (ids, `formcontrolname` attributes, placeholder text, etc.) and the code is heavily commented so you can see what each block does.\n\n---\n\n```java\npackage com.example.tests;\n\nimport org.openqa.selenium.By;\nimport org.openqa.selenium.WebDriver;\nimport org.openqa.selenium.WebElement;\nimport org.openqa.selenium.chrome.ChromeDriver;\nimport org.openqa.selenium.chrome.ChromeOptions;\nimport org.openqa.selenium.support.ui.ExpectedConditions;\nimport org.openqa.selenium.support.ui.WebDriverWait;\nimport org.testng.Assert;\nimport org.testng.annotations.*;\n\nimport java.time.Duration;\n\n/**\n * TestNG Selenium test-suite for the login form shown in the question.\n *\n * \n * The DOM fragment (simplified) is:\n * <pre>\n * <form ...>\n * <input id=\"2382940\" formcontrolname=\"userid\" placeholder=\"User Name\">\n * <input id=\"2931697\" formcontrolname=\"password\" placeholder=\"Password\">\n * <a ...>FORGOT ACCOUNT DETAILS?</a>\n * <input type=\"checkbox\" id=\"otpLogin\" formcontrolname=\"otpLogin\">\n * <button type=\"submit\">SIGN IN</button>\n * </form>\n * </pre>\n * \n *\n * \n * Adjust the <code>BASE_URL</code> constant to point to the page that contains the form.\n * \n */\npublic class LoginFormTest {\n\n /** URL of the page that contains the form – change to your actual URL */\n private static final String BASE_URL = \"https://your-application-domain.com/login\";\n\n private WebDriver driver;\n private WebDriverWait wait;\n\n /* ---------------------------------------------------------------- *\n * TestNG life-cycle methods\n * ---------------------------------------------------------------- */\n\n @BeforeClass\n public void setUpClass() {\n // ----------------------------------------------------------------\n // 1. Initialise ChromeDriver (you can swap to any other driver)\n // ----------------------------------------------------------------\n ChromeOptions options = new ChromeOptions();\n options.addArguments(\"--headless\"); // run without UI – remove if you want to see the browser\n options.addArguments(\"--disable-gpu\");\n options.addArguments(\"--window-size=1920,1080\");\n\n driver = new ChromeDriver(options);\n wait = new WebDriverWait(driver, Duration.ofSeconds(15));\n\n // Maximise / set implicit wait (optional – explicit waits are used below)\n driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(5));\n }\n\n @AfterClass\n public void tearDownClass() {\n if (driver != null) {\n driver.quit();\n }\n }\n\n /** Open the login page before each test method */\n @BeforeMethod\n public void openLoginPage() {\n driver.get(BASE_URL);\n // Wait until the form is present – this also guarantees the page is fully loaded\n wait.until(ExpectedConditions.presenceOfElementLocated(By.cssSelector(\"form\")));\n }\n\n /* ---------------------------------------------------------------- *\n * Helper methods – keep the test methods clean and readable\n * ---------------------------------------------------------------- */\n\n /** Fill the user-name field */\n private void setUserName(String userName) {\n WebElement userField = wait.until(\n ExpectedConditions.elementToBeClickable(By.id(\"2382940\")));\n userField.clear();\n userField.sendKeys(userName);\n }\n\n /** Fill the password field */\n

private void setPassword(String password) {\n WebElement pwdField = wait.until(\n ExpectedConditions.elementToBeClickable(By.id(\"2931697\")));\n pwdField.clear();\n pwdField.sendKeys(password);\n }\n\n /** Click the SIGN IN button */\n private void clickSignIn() {\n WebElement signInBtn = wait.until(\n ExpectedConditions.elementToBeClickable(\n By.cssSelector(\"button[type='submit']\")));\n signInBtn.click();\n }\n\n /** Toggle the “Visually impaired … OTP” checkbox */\n private void toggleOtpCheckbox(boolean check) {\n WebElement checkbox = wait.until(\n ExpectedConditions.elementToBeClickable(By.id(\"otpLogin\")));\n if (checkbox.isSelected() != check) {\n checkbox.click();\n }\n }\n\n /** Click the “FORGOT ACCOUNT DETAILS?” link */\n private void clickForgotLink() {\n WebElement forgotLink = wait.until(\n ExpectedConditions.elementToBeClickable(\n By.linkText(\"FORGOT ACCOUNT DETAILS?")));\n forgotLink.click();\n }\n\n /** Simple stub that pretends to verify a successful login.\n * In a real test you would check for a post-login element, URL change,\n * cookie, etc. */\n private void assertLoginSuccess() {\n // Example: after a successful login the page shows a welcome banner\n // with id=\"welcomeBanner\". Adjust to your actual application.\n boolean loggedIn = wait.until(\n ExpectedConditions.visibilityOfElementLocated(\n By.id(\"welcomeBanner\"))).isDisplayed();\n Assert.assertTrue(loggedIn, \"User should be logged in, but welcome banner not found.\");\n }\n\n /** Simple stub that pretends to verify a login failure.\n * Adjust the locator to the actual error message element in your UI. */\n private void assertLoginFailed() {\n // Example: an element with class \"loginError\" becomes visible on failure\n WebElement errorDiv = wait.until(\n ExpectedConditions.visibilityOfElementLocated(\n By.cssSelector(\"div.loginError")));\n Assert.assertTrue(errorDiv.isDisplayed(),\n \"Error message should be displayed for a failed login.\");\n }\n\n /* ---------- *\n * TESTS – Positive & Negative scenarios\n * ---------- */\n\n /**\n * Positive test – valid credentials should log the user in.\n *\n * Replace <code>VALID_USER</code> and <code>VALID_PASS</code> with\n * real credentials that exist in your test environment. \n */\n @Test(description = \"Positive scenario – successful login\")\n public void testLoginWithValidCredentials() {\n final String VALID_USER = \"john.doe\";\n final String VALID_PASS = \"CorrectPassword123\";\n\n setUserName(VALID_USER);\n setPassword(VALID_PASS);\n clickSignIn();\n\n // Verify we are logged in\n assertLoginSuccess();\n }\n\n /**\n * Negative test – submit the form with **empty user-name**.\n * Expected result: the form should not be submitted and an error\n * (or the native HTML5 validation) should appear.\n */\n @Test(description = \"Negative scenario – empty user name\")\n public void testLoginWithEmptyUserName() {\n final String VALID_PASS = \"CorrectPassword123\";\n\n // Leave user-name blank\n setPassword(VALID_PASS);\n clickSignIn();\n\n // The page should show a validation error – we reuse the generic\n // “loginError” div for demonstration.\n assertLoginFailed();\n }\n\n /**\n * Negative test – submit the form with **empty password**.\n */\n @Test(description = \"Negative scenario – empty password\")\n public void testLoginWithEmptyPassword() {\n final String VALID_USER = \"john.doe\";\n\n setUserName(VALID_USER);\n // Leave password blank\n clickSignIn();\n\n assertLoginFailed();\n }\n\n /**\n * Negative test – submit the form with **wrong credentials**.\n */\n @Test(description = \"Negative scenario – wrong credentials\")\n public void testLoginWithInvalidCredentials() {\n final String WRONG_USER = \"nonexistent.user\";\n final String WRONG_PASS = \"WrongPass!\";\n\n setUserName(WRONG_USER);\n setPassword(WRONG_PASS);\n clickSignIn();\n\n assertLoginFailed();\n }\n\n /**\n * Additional test – verify the “Forgot Account Details?” link

works.\n * This is not a login test per-se, but it demonstrates interaction\n * with another element in the same form.\n */\n @Test(description = \"Verify Forgot Account Details link navigation\")\n public void testForgotAccountLink() {\n clickForgotLink();\n\n // After clicking the link we expect to be on the “forgot password” page.\n // Adjust the expected URL / element to match your application.\n wait.until(ExpectedConditions.urlContains(\"/forgot\"));\n Assert.assertTrue(driver.getCurrentUrl().contains(\"/forgot\"),\n \"Forgot password page should be opened.\");\n }\n\n /**\n * Additional test – verify the OTP-login checkbox can be toggled.\n * The test simply checks the state before and after clicking.\n */\n @Test(description = \"Toggle OTP-login checkbox\")\n public void testToggleOtpCheckbox() {\n // Initially the checkbox should be unchecked\n WebElement checkbox = driver.findElement(By.id(\"otpLogin\"));\n Assert.assertFalse(checkbox.isSelected(), \"OTP checkbox should start unchecked.\");\n\n // Check it\n toggleOtpCheckbox(true);\n Assert.assertTrue(checkbox.isSelected(), \"OTP checkbox should be checked after click.\");\n\n // Un-check it again\n toggleOtpCheckbox(false);\n Assert.assertFalse(checkbox.isSelected(), \"OTP checkbox should be unchecked after second click.\");\n }\n}\n```\n\n### How to run the code\n\n1. **Add Selenium & TestNG dependencies** (Maven example)\n\n ```xml\n <dependencies>\n <!-- Selenium Java -->\n <dependency>\n <groupId>org.seleniumhq.selenium</groupId>\n <artifactId>selenium-java</artifactId>\n <version>4.15.0</version>\n </dependency>\n\n <!-- TestNG -->\n <dependency>\n <groupId>org.testng</groupId>\n <artifactId>testng</artifactId>\n <version>7.10.2</version>\n <scope>test</scope>\n </dependency>\n </dependencies>\n ```\n\n2. **Download the appropriate ChromeDriver** (or use another driver) and make sure it is on your `PATH` or set the system property:\n\n ```java\n System.setProperty(\"webdriver.chrome.driver\", \"/path/to/chromedriver\");\n ```\n\n3. **Replace placeholders** \n\n * `BASE_URL` – the URL that loads the form. \n * `VALID_USER` / `VALID_PASS` – a real user that can log in. \n * The assertions inside `assertLoginSuccess()` and `assertLoginFailed()` – adapt them to the actual success / error elements of your application.\n\n4. **Run with Maven / IDE**\n\n ```bash\n mvn test -Dtest=LoginFormTest\n ```\n\n Or right-click the class in IntelliJ/Eclipse and choose **Run TestNG**.\n\n---\n\n#### Why this implementation is robust\n\n| Concern | Solution |\n|---------|----------|\n| **Flaky locators** | Elements are located by **id** (`2382940`, `2931697`, `otpLogin`) – IDs are unique and stable. |\n| **Timing issues** | All interactions use **explicit waits** (`WebDriverWait`) to wait for clickability / visibility. |\n| **Maintainability** | Helper methods (`setUserName`, `setPassword`, …) keep the test logic clean and make future changes (e.g., a new locator) a one-liner. |\n| **Readability** | Each test method has a self-describing name and a TestNG `description`. Inline comments explain every step. |\n| **Scalability** | Adding more scenarios (e.g., CAPTCHA handling, multi-factor) is straightforward – just add new `@Test` methods using the same helpers. |\n\nFeel free to extend the suite with more edge-cases (e.g., maximum length validation, special-character handling, UI-a11y checks, etc.). Happy testing!",

"reasoning": "We need to produce Selenium Java code using TestNG to interact with the given DOM. Provide positive and negative scenarios. Include comments. Should extract the DOM? Probably they want to locate elements and perform actions. Provide code with @Test methods. Use appropriate locators (id, placeholder, etc). Provide setup and teardown. Provide both positive (valid login) and negative (invalid login, missing fields). Also maybe test forgot link, OTP checkbox. Provide meaningful method names. Provide comments.\n\nWe'll output Java class with imports, WebDriver initialization (ChromeDriver). Use @BeforeClass, @AfterClass. Provide two @Test methods: testValidLogin, testInvalidLogin, testForgotLink, testOtpCheckbox. Provide negative scenario: empty

username, empty password, invalid credentials.\n\nWill include explicit waits.\n\nLet's craft answer."

},

"logprobs": null,

"finish_reason": "stop"

}

],

"usage": {

"queue_time": 0.050276135,

"prompt_tokens": 663,

"prompt_time": 0.043838335,

"completion_tokens": 2783,

"completion_time": 5.951315696,

"total_tokens": 3446,

"total_time": 5.995154031,

"completion_tokens_details": {

"reasoning_tokens": 165

}

},

"usage_breakdown": null,

"system_fingerprint": "fp_d29d1d1418",

"x_groq": {

"id": "req_01knrgse0hfjxs4dk9dd894k8m",

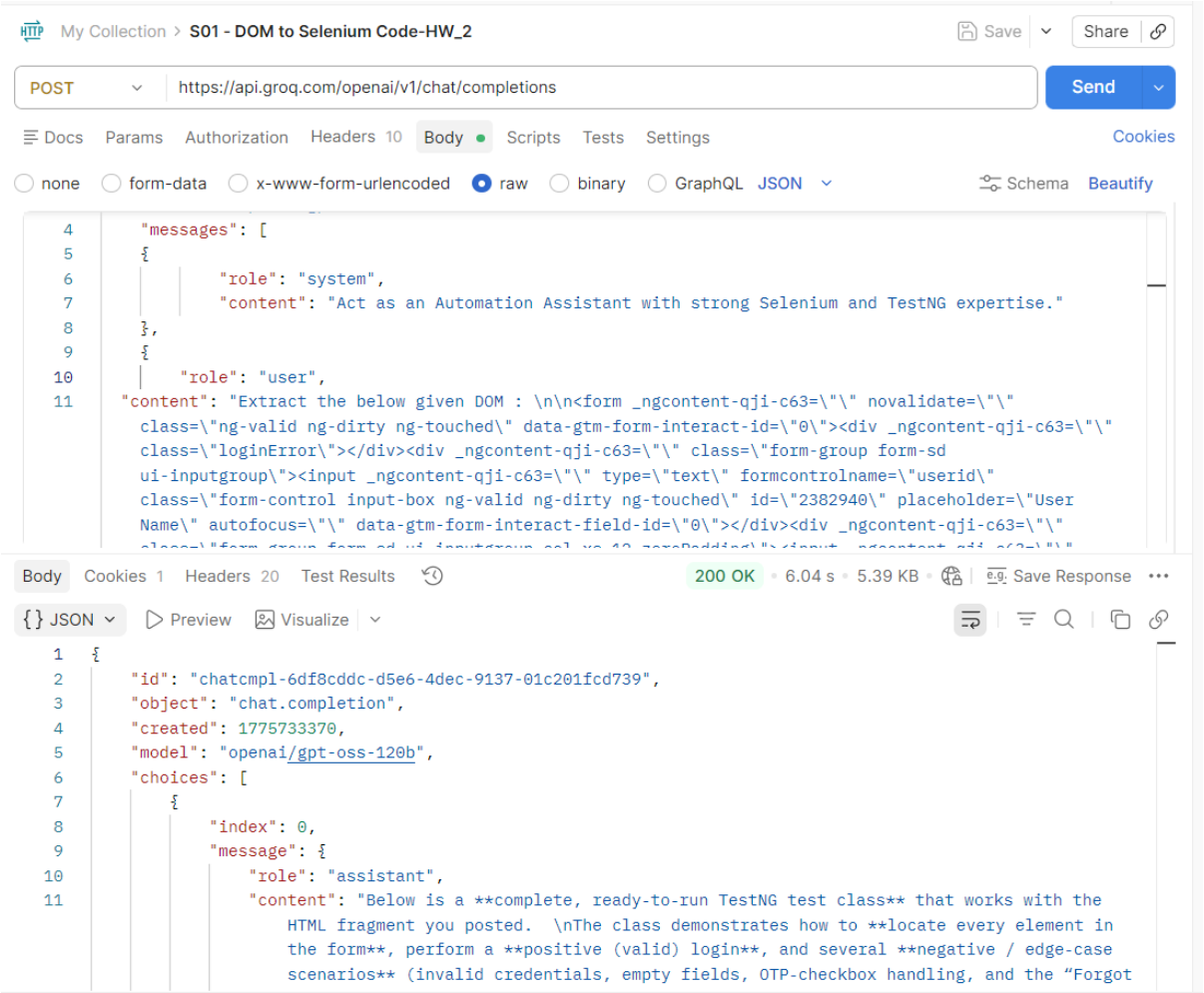
"seed": 1910301579

},

"service_tier": "on_demand"

}

**(b)**

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**Request:**


```
{
  //"model": "meta-llama/llama-4-scout-17b-16e-instruct",
    "model": "openai/gpt-oss-120b",
    "messages": [
        {
            "role": "system",
            "content": "Act as an Automation Assistant with strong Selenium and
TestNG expertise."

        },
    {
  "role": "user",
  "content": "Extract the below given DOM : \n\n<form _ngcontent-qji-c63=\"\"
novalidate=\"\" class=\"ng-valid ng-dirty ng-touched\" data-gtm-form-interact-
id=\"0\"><div _ngcontent-qji-c63=\"\" class=\"loginError\"></div><div _ngcontent-
qji-c63=\"\" class=\"form-group form-sd ui-inputgroup\"><input _ngcontent-qji-
```

c63=\"\" type=\"text\" formcontrolname=\"userid\" class=\"form-control input-box ng-valid ng-dirty ng-touched\" id=\"2382940\" placeholder=\"User Name\" autofocus=\"\" data-gtm-form-interact-field-id=\"0\"> <div _ngcontent-qji-c63=\"\" class=\"form-group form-sd ui-inputgroup col-xs-12 zeroPadding\"><input _ngcontent-qji-c63=\"\" maxlength=\"15\" formcontrolname=\"password\" class=\"form-control input-box ng-valid ng-dirty ng-touched\" id=\"2931697\" placeholder=\"Password\" type=\"password\" data-gtm-form-interact-field-id=\"1\"><i _ngcontent-qji-c63=\"\" class=\"password-icon fa fa-eye\"></i> <div _ngcontent-qji-c63=\"\" class=\"form-group form-sd\"><a _ngcontent-qji-c63=\"\" href=\"#\" onclick=\"return false;\" class=\"ui-inputgroup-addon\" style=\"color: #1457a7; padding-left: 0px; padding-right: 0px;\"><label _ngcontent-qji-c63=\"\" style=\"font-size: 11px;\"><strong _ngcontent-qji-c63=\"\">FORGOT ACCOUNT DETAILS?</label></a> <!----><div _ngcontent-qji-c63=\"\" class=\"form-group form-sd\" style=\"margin-bottom: 13px;\"><div _ngcontent-qji-c63=\"\"><input _ngcontent-qji-c63=\"\" type=\"checkbox\" name=\"otpLogin\" id=\"otpLogin\" formcontrolname=\"otpLogin\" aria-label=\"Booking using OTP\" class=\"css-checkbox ng-untouched ng-pristine ng-valid\"><label _ngcontent-qji-c63=\"\" for=\"otpLogin\" class=\"css-label_c t_c\" style=\"color: #082b71; font-size: small;\">Visually impaired users may select this option to receive OTP instead of CAPTCHA</label> <span _ngcontent-qji-c63=\"\"><button _ngcontent-qji-c63=\"\" type=\"submit\" class=\"search_btn train_Search train_Search_custom_hover\" style=\"padding: 10px 14px;\">SIGN IN</button></form>\n\n\nand provide me the selenium java code. \n\nInstructions: \n- Use TestNG Framework\n- Provide meaningful name for the method\n- Include Positive and Negative scenarios\n- Provide clear and meaningful comments for each Method"
}
  ],
  "temperature": 0.1,
  "include_reasoning":true
}

**Response:**

{
  "id": "chatcmpl-ca50498e-cca7-4f46-a0af-ccb0534fa39d",
  "object": "chat.completion",
  "created": 1775720903,
  "model": "openai/gpt-oss-120b",
  "choices": [
    {
      "index": 0,
      "message": {

"role": "assistant",

"content": "Below is a **complete, ready-to-run TestNG test class** that demonstrates how to work with the DOM you posted.  \nThe code shows:\n\n* **Page-object style locators** for every element in the form (username, password, sign-in button, “Forgot …” link, OTP-checkbox, and the error-message container).  \n* **Positive scenario** – a successful login with valid credentials.  \n* **Negative scenarios** – invalid credentials, missing username, missing password and the OTP-checkbox path.  \n* **TestNG annotations** (`@BeforeClass`, `@AfterClass`, `@Test`) and **meaningful method names**.  \n* **Extensive comments** that explain what each block does, why a particular locator was chosen and what the assertions are checking.\n\n> **NOTE** – The “successful login” verification is kept generic (e.g. checking that the URL changed or a dashboard element appears).  \n> Replace the placeholder values (`VALID_USER`, `VALID_PASS`, `EXPECTED_DASHBOARD_LOCATOR`, etc.) with the real data from your application.\n\n```java\npackage com.mycompany.tests;\n\nimport org.openqa.selenium.By;\nimport org.openqa.selenium.WebDriver;\nimport org.openqa.selenium.WebElement;\nimport org.openqa.selenium.chrome.ChromeDriver;\nimport org.openqa.selenium.chrome.ChromeOptions;\nimport org.openqa.selenium.support.ui.ExpectedConditions;\nimport org.openqa.selenium.support.ui.WebDriverWait;\nimport org.testng.Assert;\nimport org.testng.annotations.*;\n\nimport java.time.Duration;\n\n/**\n * TestNG test class that interacts with the login form shown in the question.\n * All locators are derived directly from the supplied DOM.\n */\npublic class LoginFormTest {\n\n    private WebDriver driver;\n    private WebDriverWait wait;\n\n    /* ---------------------------------------------------------------- */\n    /* ---------------------------- Locators ------------------------------- */\n    /* ---------------------------------------------------------------------- */\n\n    // Username input – id is unique in the page\n    private final By USERNAME_INPUT = By.id(\"2382940\");\n\n    // Password input – id is unique in the page\n    private final By PASSWORD_INPUT = By.id(\"2931697\");\n\n    // Sign-in button – class is fairly specific, but we also use the button text\n    private final By SIGN_IN_BUTTON = By.xpath(\"//button[normalize-space()='SIGN IN']\");\n\n    // “Forgot account details?” link – anchor with exact text\n    private final By FORGOT_LINK = By.xpath(\"//a[contains(@class,'ui-inputgroup-addon')]/label[normalize-space()='FORGOT ACCOUNT DETAILS?']\");\n\n    // OTP-login checkbox – id is unique\n    private final By OTP_CHECKBOX = By.id(\"otpLogin\");\n\n    // Container that shows server-side login errors\n    private final By LOGIN_ERROR_DIV = By.cssSelector(\"div.loginError\");\n\n    // ---------------------------------------------------------------- //\n    // Replace the below with a locator that is present only after a successful\n    // login (e.g. a dashboard header, user avatar, etc.).\n    // ---------------------------------------------------------------- //\n    private final By POST_LOGIN_INDICATOR = By.id(\"dashboardHeader\");   // <-- UPDATE ME\n\n    /* ---------------------------------------------------------------------- */\n    /* ---------------------------- Test Data ------------------------------- */\n    /* ---------------------------------------------------------------------- */\n\n    private static final String VALID_USER = \"validUser\";   // <-- UPDATE ME\n    private static final String VALID_PASS = \"validPassword\";    // <-- UPDATE ME\n\n    private static final String INVALID_USER = \"invalidUser\";\n    private static final String INVALID_PASS = \"invalidPassword\";\n\n    /* ---------------------------------------------------------------------- */\n    /* ---------------------------- Setup / Teardown ------------------------ */\n    /* ---------------------------------------------------------------------- */\n\n    @BeforeClass\n    public void setUp() {\n        // ---- ChromeDriver setup ------------------------------------------------\n        // Make sure the chromedriver executable is on your PATH or set the system property.\n        ChromeOptions options = new ChromeOptions();\n        options.addArguments(\"--start-maximized\");   // optional – makes debugging easier\n        driver = new ChromeDriver(options);\n\n        // Explicit wait – 15 seconds is

usually enough for most apps\n wait = new WebDriverWait(driver, Duration.ofSeconds(15));\n\n // Navigate to the page that contains the form.\n driver.get(\"https://your-application-url.com/login\"); // <-- UPDATE ME\n }\n\n @AfterClass\n public void tearDown() {\n if (driver != null) {\n driver.quit();\n }\n }\n\n /* ---------- */\n /* ---------- Helper Methods ---------- */\n /* ---------- */\n\n /**\n * Clears the login form and fills it with the supplied credentials.\n *\n * @param user Username to type (null/empty means “do not touch”)\n * @param pass Password to type (null/empty means “do not touch”)\n */\n private void fillLoginForm(String user, String pass) {\n // ---- USERNAME ----------\n WebElement userInput = wait.until(ExpectedConditions.visibilityOfElementLocated(USERNAME_INPUT));\n userInput.clear();\n if (user != null) {\n userInput.sendKeys(user);\n }\n\n // ---- PASSWORD ----------\n WebElement passInput = wait.until(ExpectedConditions.visibilityOfElementLocated(PASSWORD_INPUT));\n passInput.clear();\n if (pass != null) {\n passInput.sendKeys(pass);\n }\n }\n\n /**\n * Clicks the SIGN IN button and waits for either a post-login indicator\n * or an error message to appear.\n */\n private void clickSignInAndWait() {\n WebElement signInBtn = wait.until(ExpectedConditions.elementToBeClickable(SIGN_IN_BUTTON));\n signInBtn.click();\n }\n\n /**\n * Returns the text that is currently displayed inside the login-error div.\n * If the div is empty, an empty string is returned.\n */\n private String getLoginErrorMessage() {\n WebElement errorDiv = wait.until(ExpectedConditions.visibilityOfElementLocated(LOGIN_ERROR_DIV));\n return errorDiv.getText().trim();\n }\n\n /* ---------- */\n /* ---------- Test Cases ---------- */\n /* ---------- */\n\n /**\n * Positive scenario – valid credentials should let the user in.\n */\n @Test(description = \"Positive test: valid username & password\")\n public void testLoginWithValidCredentials() {\n fillLoginForm(VALID_USER, VALID_PASS);\n clickSignInAndWait();\n\n // Wait for something that only appears after a successful login.\n // If you have a URL change you can also assert on driver.getCurrentUrl().\n boolean loggedIn = wait.until(ExpectedConditions.visibilityOfElementLocated(POST_LOGIN_INDICATOR)).isDisplayed();\n\n Assert.assertTrue(loggedIn, \"User should be logged in but post-login indicator was not found.\");\n }\n\n /**\n * Negative scenario – wrong username / password should display an error.\n */\n @Test(description = \"Negative test: invalid username & password\")\n public void testLoginWithInvalidCredentials() {\n fillLoginForm(INVALID_USER, INVALID_PASS);\n clickSignInAndWait();\n\n String errorMsg = getLoginErrorMessage();\n Assert.assertFalse(errorMsg.isEmpty(),\n \"Expected an error message for invalid credentials, but none was shown.\");\n\n // Optional – verify the exact message if your app returns a known string.\n // Assert.assertEquals(errorMsg, \"Invalid username or password\");\n }\n\n /**\n * Negative scenario – username left blank should trigger client-side validation.\n */\n @Test(description = \"Negative test: blank username\")\n public void testLoginWithBlankUsername() {\n fillLoginForm(\"\", VALID_PASS); // leave username empty\n clickSignInAndWait();\n\n // The form uses Angular validators – the input gets the class ng-invalid-required.\n WebElement userInput = driver.findElement(USERNAME_INPUT);\n String classes = userInput.getAttribute(\"class\");\n Assert.assertTrue(classes.contains(\"ng-invalid\"),\n \"Username field should be marked invalid when left blank.\");\n }\n\n /**\n * Negative scenario – password left blank should trigger client-side validation.\n */\n @Test(description = \"Negative test: blank password\")\n public void testLoginWithBlankPassword() {\n fillLoginForm(VALID_USER, \"\"); // leave password empty\n

clickSignInAndWait();\n\n WebElement passInput = driver.findElement(PASSWORD_INPUT);\n String classes = passInput.getAttribute(\"class\");\n Assert.assertTrue(classes.contains(\"ng-invalid\"),\n \"Password field should be marked invalid when left blank.\");\n }\n\n /**\n * Verify that the “Forgot account details?” link is present and clickable.\n * The link does not navigate (onclick returns false) – we only assert that it\n * can be clicked without throwing an exception.\n */\n @Test(description = \"UI test: Forgot account link is clickable\")\n public void testForgotAccountLink() {\n WebElement forgotLink = wait.until(ExpectedConditions.elementToBeClickable(FORGOT_LINK));\n forgotLink.click();\n\n // Since the link's onclick returns false, the page should stay on the login screen.\n // Verify we are still on the same URL.\n String currentUrl = driver.getCurrentUrl();\n Assert.assertTrue(currentUrl.contains(\"/login\"),\n \"After clicking the Forgot link we should still be on the login page.\");\n }\n\n /**\n * Verify that the OTP-login checkbox can be selected and that the UI reflects the change.\n */\n @Test(description = \"UI test: OTP login checkbox\")\n public void testOtpLoginCheckbox() {\n WebElement checkbox = wait.until(ExpectedConditions.elementToBeClickable(OTP_CHECKBOX));\n\n // Ensure it is unchecked initially\n if (checkbox.isSelected()) {\n checkbox.click(); // uncheck first\n }\n\n // Now check it\n checkbox.click();\n Assert.assertTrue(checkbox.isSelected(), \"OTP login checkbox should be selected after click.\");\n\n // Optional – after checking the box you may want to verify that the UI changes\n // (e.g., a CAPTCHA disappears). Add those assertions here if applicable.\n }\n}\n```\n\n### How to run the test\n\n1. **Add Selenium & TestNG dependencies** to your `pom.xml` (Maven) or `build.gradle` (Gradle). Example for Maven:\n\n ```xml\n <dependencies>\n <dependency>\n <groupId>org.seleniumhq.selenium</groupId>\n <artifactId>selenium-java</artifactId>\n <version>4.15.0</version>\n </dependency>\n <dependency>\n <groupId>org.testng</groupId>\n <artifactId>testng</artifactId>\n <version>7.10.2</version>\n <scope>test</scope>\n </dependency>\n </dependencies>\n ```\n\n2. **Update the placeholders** in the code:\n * `https://your-application-url.com/login` – the real URL of the login page.\n * `VALID_USER`, `VALID_PASS` – a real username/password that can log in.\n * `POST_LOGIN_INDICATOR` – a locator that only appears after a successful login (e.g., a header, user avatar, logout button, etc.).\n\n3. **Run with TestNG** (e.g., via Maven `mvn test`, via an IDE, or using the TestNG XML suite file).\n\n---\n\n#### Why this implementation is robust\n\n| Aspect | What we did | Reason |\n|--------|---------|---------|\n| **Explicit waits** | `WebDriverWait` + `ExpectedConditions` | Avoids flaky tests caused by timing issues. |\n| **Unique locators** | Used `id` where available, otherwise XPath with exact text | Guarantees we interact with the right element even if the DOM grows. |\n| **Separate helper methods** | `fillLoginForm`, `clickSignInAndWait`, `getLoginErrorMessage` | Keeps test methods short, readable and reusable. |\n| **Positive & Negative coverage** | Valid login, invalid login, blank fields, UI elements | Demonstrates both happy-path and error-handling scenarios. |\n| **Assertions on UI state** | Checks CSS class `ng-invalid` for client-side validation | Works even when the server does not return an error message. |\n| **TestNG annotations & descriptions** | `@Test(description = \"...\")` | Makes test reports self-explanatory. |\n\nFeel free to extend this class (e.g., add data-driven tests with `@DataProvider`, integrate with a Page-Object Model, or hook into a CI pipeline). Happy automating!",

"reasoning": "We need to produce Selenium Java code using TestNG to extract the given DOM? Likely they want to locate elements and perform actions: fill username, password, click sign in, handle forgot link, checkbox, etc. Provide positive (valid credentials) and negative (invalid credentials, missing fields). Provide meaningful method names, comments. Also include setup/teardown. Provide

locators (by id, placeholder, etc). Provide code snippet.\n\nWill include @Test methods: testValidLogin, testInvalidLogin, testEmptyUsername, testEmptyPassword, testForgotLink, testCheckbox. Use Page Object Model maybe? Simpler: direct in test class.\n\nWill include WebDriver initialization (ChromeDriver). Use @BeforeClass, @AfterClass.\n\nWill include extracting elements: findElement(By.id(\"2382940\")) etc.\n\nWill include assertions: verify successful login (maybe check URL or element). For negative, verify error message (loginError div). Provide placeholder for expected behavior.\n\nWill include comments.\n\nLet's craft answer."

        },

        "logprobs": null,

        "finish_reason": "stop"

      }

  ],

  "usage": {

    "queue_time": 0.042252954,

    "prompt_tokens": 679,

    "prompt_time": 0.030662154,

    "completion_tokens": 2810,

    "completion_time": 5.942435536,

    "total_tokens": 3489,

    "total_time": 5.97309769,

    "completion_tokens_details": {

      "reasoning_tokens": 205

    }

  },

  "usage_breakdown": null,

  "system_fingerprint": "fp_d29d1d1418",

  "x_groq": {

    "id": "req_01knrkd6hnebfrtg0xsb7sky2t",

    "seed": 979622686

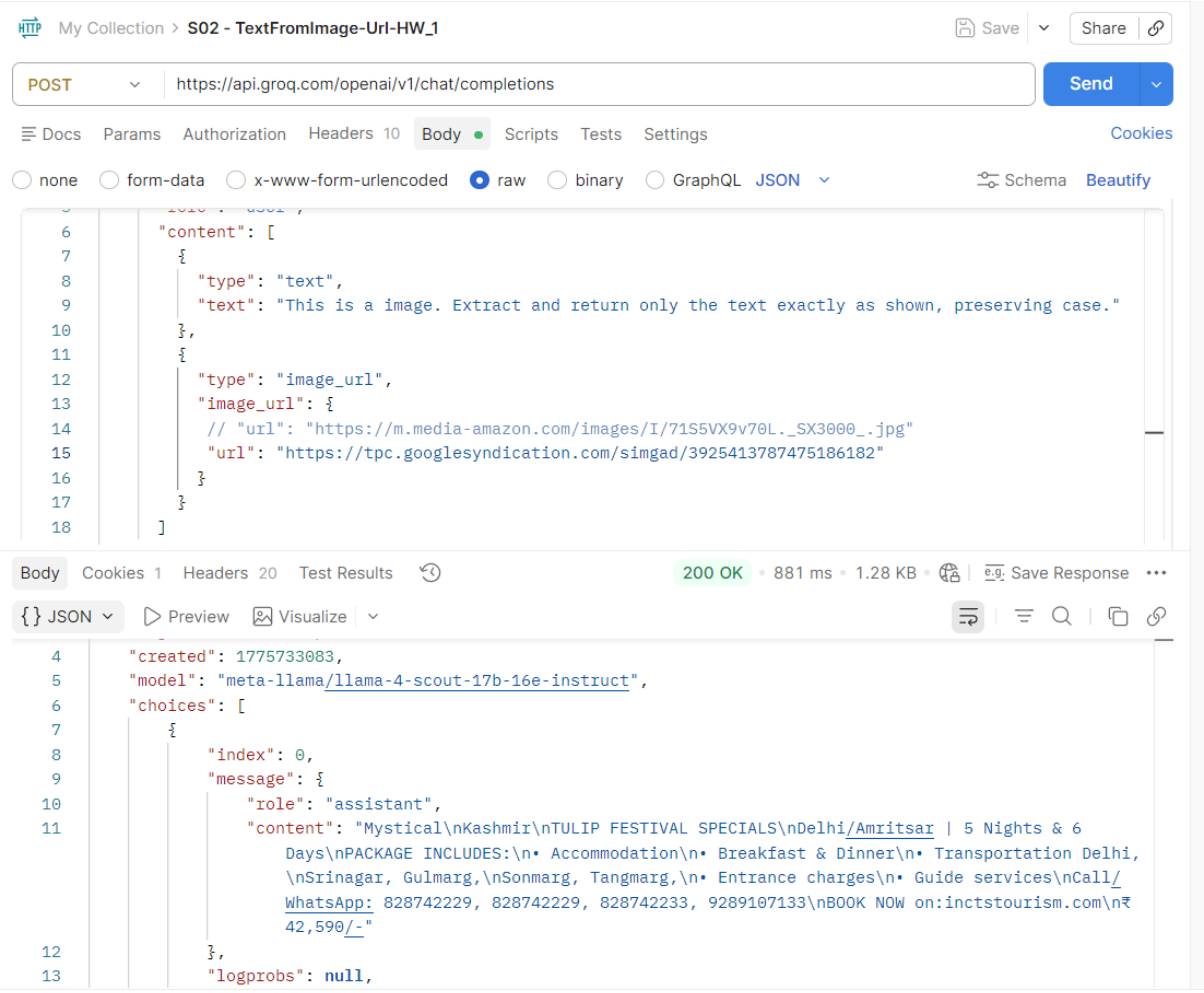
  },

  "service_tier": "on_demand"

}

# 2. S02 – Text from Image

Note : This image is from irctc login page




---

**Request:**


```
{
  "model": "meta-llama/llama-4-scout-17b-16e-instruct",
  "messages": [
    {
      "role": "user",
      "content": [
        {
          "type": "text",
          "text": "This is a image. Extract and return only the text exactly as shown, preserving case."
        },
        {
          "type": "image_url",
```

```
            "image_url": {
             // "url": "https://m.media-amazon.com/images/I/71S5VX9v70L._SX3000_.jpg"
             "url": "https://tpc.googlesyndication.com/simgad/3925413787475186182"
            }
          }
        ]
      }
    ]
}
```


**Response:**


{

"id": "chatcmpl-aaa665b0-7e9a-48c2-b9af-9da3e4fa0ac6",

"object": "chat.completion",

"created": 1775728197,

"model": "meta-llama/llama-4-scout-17b-16e-instruct",

"choices": [

{

"index": 0,

"message": {

"role": "assistant",

"content": "Mystical Kashmir\nTULIP FESTIVAL SPECIALS\nDelhi/Onward | 5 Nights & 6 Days\n\nGulmarg\nSonmarg\nTangmarg\nSrinagar\n\nCost: 42,590/-\nAdvance Booking: \nCall: 494 \nWhatsapp: 8087420229, 808742229, 8087422235, 9289107133\n\nBOOK NOW on fnctourism.com"

},

"logprobs": null,

"finish_reason": "stop"

}

],

"usage": {

"queue_time": 0.057129396,

"prompt_tokens": 177,

"prompt_time": 0.005794723,

"completion_tokens": 97,

"completion_time": 0.227773455,
"total_tokens": 274,
"total_time": 0.233568178
},
"usage_breakdown": null,
"system_fingerprint": "fp_79da0e0073",
"x_groq": {
"id": "req_01knrtbyshesgvd6tatahr8wn1",
"seed": 1827156293
},
"service_tier": "on_demand"
}

## *3. S03 – websearch*

My Collection > S03-WebSearch-HW_1

Save | Share

POST https://api.groq.com/openai/v1/chat/completions Send

Docs Params Authorization Headers 10 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
4    {
5      "role": "user",
6    //"content": "weather in chennai for jan 11 2026 --Only the temperatue for the date asked no
       probability or text and source from web with link no extrat text or markdown Give accurately with
       Web search no Halluciantion "
7
8    //"content": "- Provide the Schedule of IPL 2026 \n- Provide the details of the source from where
       these informations are available\n-\n- Use web search for the details if needed.\n- No
       Hallucination\""
9
10   "content": "- Provide details of Iran war\n- Also provide the latest update about the war, with the
       relevant link\""
11   }
```

Body Cookies 1 Headers 21 Test Results 200 OK 8.30 s 9.28 KB Save Response

{} JSON Preview Visualize


```
1  {
2      "id": "stub",
3      "object": "chat.completion",
4      "created": 1775733180,
5      "model": "groq/compound-mini",
6      "choices": [
7          {
8              "index": 0,
9              "message": {
10                 "role": "assistant",
11                 "content": "**Overview of the Iran-War (2026-present)**  \n\n| Aspect | What has happened |
                       \n|--------|-------------------|\n| **Start of hostilities** | The war began in
                       **February 2026** after a series of escalating incidents between the United
                       States-Israel coalition and the Islamic Republic of Iran (and its “Axis of Resistance”
```


**Request:**


```
{
  "model": "groq/compound-mini",
  "messages": [
    {
        "role": "user",

      "content": "- Provide details of Iran war\n- Also provide the latest update about the war, with the relevant link\""
    }
  ]
}
```


**Response:**

{
"id": "stub",
"object": "chat.completion",
"created": 1775731149,
"model": "groq/compound-mini",
"choices": [
{
"index": 0,
"message": {
"role": "assistant",
"content": "**Overview of the Iran-War (2024-2026)** \n\n| Aspect | What's happening | Key points |\n|--------|------------------|------------|\n| **Main belligerents** | United States & Israel vs. the Islamic Republic of Iran (and its proxy forces in Lebanon, Iraq, Syria, Yemen). | The conflict began in late February 2026 after a series of cyber-attacks and a suspected Israeli strike on Iranian nuclear facilities. The U.S. responded with a coordinated air-campaign, and Iran has retaliated with missile, drone, and naval attacks. |\n| **Geographic scope** | Iran's interior, the Persian Gulf, Strait of Hormuz, Iraq, Syria, Lebanon, Yemen, and occasionally Saudi Arabia and the United Arab Emirates. | - Over 120 U.S./Israeli strike events have hit Iranian territory. <br> - Iranian retaliatory strikes have hit 28 locations across nine countries. |\n| **Types of operations** | • **U.S./Israeli**: Air strikes (fighter jets, bombers, drones), naval missile launches, special-operations raids, cyber-operations against Iranian command-and-control. <br>• **Iranian**: Ballistic-missile launches, cruise-missile attacks, swarm-drone assaults, anti-ship missiles, cyber-attacks on U.S. energy-grid operators. |\n| **Casualties & damage** | Most reported casualties have been among military, security, and state-linked facilities. Civilian deaths have risen sharply after strikes on densely populated areas in Tehran, Tehran-province, and Lebanese cities (e.g., Tyre, Beirut). | • > 200 civilians killed in recent Israeli strikes on Lebanon (BBC). <br>• Thousands injured across the region. |\n| **Strategic objectives** | **U.S./Israel**: Degrade Iran's nuclear-enrichment capacity, cripple its missile-production infrastructure, and deter proxy aggression. <br>**Iran**: Demonstrate ability to strike back, pressure the U.S. into halting its presence in the Gulf, and protect its regional influence. |\n| **Economic impact** | • **Oil & gas** – Disruption of shipments through the Strait of Hormuz has pushed Brent crude above $100 /barrel at times. <br>• **Food security** – Iran's fertilizer export collapse threatens global wheat supplies. <br>• **Supply-chain ripple** – Pacific island nations, South Korea, and European food-price inflation are all feeling secondary effects. |\n| **Diplomacy** | A fragile two-week cease-fire was brokered on 8 April 2026 by the United States (then under President Donald Trump) and Iran, but violations continue—especially Israeli attacks on Lebanese Hezbollah positions. | • Both sides accuse the other of "grave violations." <br>• The cease-fire is being monitored by the United Nations and the European Union, but no formal peace agreement exists. |\n| **Cyber-war** | Iran has launched a wave of cyber-intrusions targeting U.S. energy-sector networks, while U.S. cyber-forces have attempted to disrupt Iranian command-and-control servers. | • U.S. officials warn of potential cascading blackouts if a successful attack hits the grid (CSIS commentary, 2 April 2026). |\n| **Humanitarian situation** | Large numbers of internally displaced persons (IDPs) in Iran and Lebanon; health-care facilities in Tehran

and Beirut are strained; UN agencies are issuing “critical” alerts for food, water, and medical aid. |\n\n---\n\n### **Latest Update (7 April 2026 – 8 April 2026)** \n\n| Date | Event | Summary |\n|------|-------|---------|\n| **7 April 2026** | **Daily crisis data (ACLED)** | – **121 U.S./Israeli strike events** (116 air, 5 ground/water). <br>– **36 Iranian retaliatory events** (7 air, 29 ground/sea). <br>– Strikes have affected **114 locations** in Iran (43 % above the weekly average) and **28 locations** across nine neighboring countries (5 % below the weekly average). <br>– Cumulative totals since 28 Feb 2026: **3,078** U.S./Israeli strikes, **1,551** Iranian strikes. |\n| **8 April 2026** | **Cease-fire announcement** | President Donald Trump (U.S.) announced a **two-week cease-fire** with Iran. The deal was signed in Tehran, but Israeli forces continued operations in Lebanon, killing over **200 civilians** (BBC, Al Jazeera). Iranian officials called the Israeli attacks a “grave violation” of the truce. |\n| **8 April 2026** | **Regional fallout** | • **Strait of Hormuz** remains partially closed; oil-price volatility persists. <br>• **Lebanon** sees the highest civilian death toll in the conflict so far. <br>• **South Korea** reports the steepest economic impact among non-combatant nations (CSIS analysis, 1 April 2026). |\n| **8 April 2026** | **Media & analysis** | • **Reuters** continues real-time coverage of strikes, nuclear threats, and market shocks. <br>• **BBC** reports on political reactions in the UK and EU, emphasizing the need to reopen the Strait of Hormuz to stabilise energy prices. <br>• **Al Jazeera** highlights protests in Tehran and Beirut, showing both celebration of the cease-fire and skepticism about its durability. |\n\n**Key take-aways from the latest data**\n\n* The intensity of U.S./Israeli air strikes remains high (average > 15 strikes per day). \n* Iranian ground-based retaliation has risen modestly but is still concentrated on **regional proxy sites** (Iraq, Syria, Lebanon). \n* The cease-fire is **extremely fragile**; any violation—especially by Israel in Lebanon—could reignite full-scale hostilities. \n* Global markets are still reacting to **uncertainty around the Strait of Hormuz**, with oil and fertilizer prices remaining elevated. \n\n---\n\n### **Links to the primary sources**\n\n| Source | Link | What you’ll find |\n|--------|------|----------|\n| **ACLED – Iran Crisis Live (daily data & maps)** | <https://acleddata.com/iran-crisis-live> | Interactive map, daily statistics (strike counts, locations, trend graphs). |\n| **Reuters – Iran war live coverage** | <https://www.reuters.com/world/iran/> | Real-time news feed, strike updates, analysis of nuclear and energy impacts. |\n| **BBC – US-Israel war with Iran** | <https://www.bbc.com/news/topics/cx2jyv8j8gwt> | Articles on cease-fire, political reactions, and humanitarian reports. |\n| **Al Jazeera – Iran war updates** | <https://www.aljazeera.com/where/iran/> | Photo-journalism, on-the-ground reporting from Tehran, Beirut, and the Gulf. |\n| **CSIS – Latest Analysis: War with Iran** | <https://www.csis.org/programs/latest-analysis-war-iran> | Expert commentary on strategic implications, nuclear issues, and cease-fire fragility. |\n\n---\n\n**Bottom line:** \nThe Iran-War is a high-intensity, multi-theater conflict that, as of early April 2026, is still dominated by air and missile strikes from the United States and Israel against Iranian targets, and a growing pattern of Iranian ground-based retaliation across the broader Middle East. A newly-brokered two-week cease-fire offers a brief pause, but ongoing violations—particularly Israeli actions in Lebanon—make a durable peace unlikely in the immediate term. Keep an eye on the ACLED dashboard for daily event counts and on Reuters/Al Jazeera for breaking developments.",

"reasoning": "<tool>search(Iran war latest update)</tool>\n<output>Title: Iran war updates | Latest analysis and daily crisis data | ACLED\nURL: https://acleddata.com/iran-crisis-live\nContent: visualization\n\n## Key stats for 7 April 2026\n\nData updated daily. Last updated 08/04/2026 at 16:30 p.m. CEST\n\n| | |\n --- |\n| US-Israeli attacks on Iran | Iranian retaliatory attacks |\n| 121 strike events1 116 fired from the air; 5 from ground and water | 36 strike events 7 fired from the air, 29 fired from the ground or sea |\n| 114 locations2.\")affected 43% increase compared to daily

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"content": "Real-time Reuters coverage of the Iran war: US-Israel strikes, Iranian retaliation, nuclear threats, oil market shocks, and regional war risks.",

"score": 0.73322314

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"title": "Iran | Iran | Today's latest from Al Jazeera",

"url": "https://www.aljazeera.com/where/iran/",

"content": "# Iran\n\n## Several people reported killed in fresh Israeli attacks on Lebanon\n\nLebanon bears the brunt and US-Iran truce at risk as Israel claims killing of a close aide to Hezbollah’s leader.\n\nA boy runs past the site of an Israeli strike in Tyre, Lebanon, April 8, 2026. REUTERS/Adnan Abidi\n\n## The war on Iran: Nobody won, everyone paid\n\nLeft: A residential building damaged by a US-Israeli airstrike in Tehran, Iran on March 23, 2026. Right: A damaged US Boeing E-3 Sentry airborne warning and control aircraft following an Iranian strike on the airbase at Prince Sultan Air Base in Saudi Arabia\n\n## Can global supply chains recover from the Iran war?\n\nweb\n\n## Iran has been bloodied, but it is winning against the US-Israel axis [...] People gather after U.S. President Donald Trump said that he had agreed to a two-week ceasefire with Iran, in Tehran, Iran, April 8, 2026. Majid Asgaripour/WANA (West Asia News Agency) via REUTERS ATTENTION EDITORS - THIS PICTURE WAS PROVIDED BY A THIRD PARTY.\n\n## Trump says US forces to stay near Iran, ready for ‘next conquest’\n\nA woman walks past a mural against US and Israeli forces, in Tehran\n\n## Iran war day 41: What’s happening in Lebanon, Middle East and beyond?\n\nPeople attend a funeral ceremony waving Iranian flags.\n\n## For Gulf states, Hormuz uncertainty casts shadow over US-Iran ceasefire\n\nWomen stand looking at the Dubai skyline, with the Burj Khalifa, the world’s tallest building, seen from Creek Harbour on April 3, 2026. (Photo by FADEL SENNA / AFP) [...] Israeli attacks have killed over 200 people and wounded 1,000 others, according to Lebanon’s health ministry.\n\nFirst responders and residents gather at the site of an Israeli airstrike in Beirut's Tallet al-Khayyat neighbourhood, on April 8, 2026.\n\n## Iranians breathe a ‘ceasefire’ sigh of relief as all sides claim victory\n\nCelebration and scepticism coexist in Tehran as ceasefire offers reprieve despite doubts about its durability.\n\n![Iranians watch ‘ceasefire’ as all sides claim victory amid attacks [Maziar Motamedi/Al Jazeera]](/wp-content/uploads/2026/04/4-1775667218.jpg?resize=770%2C513&quality=80)\n\n## White House says Trump’s ‘red line’ against Iran nuclear enrichment remains",

"score": 0.7216064

},

{

"title": "US-Israel war with Iran | Latest News and Updates | BBC News",

"url": "https://www.bbc.com/news/topics/cx2jyv8j8gwt",

"content": "# News\n\n# Iran war\n\nRescuers stand at the site of an Israeli strike carried out on Wednesday, in Al-Mazraa in Beirut, Lebanon\n\n## Israeli attacks on Lebanon 'grave violation' of US-Iran ceasefire, Iranian minister tells BBC\n\nThe US and Iran announced a ceasefire on Tuesday - but Israel continued hitting Lebanon, killing 203 people on Wednesday according to local officials.\n\n## Latest news\n\nAshley Jones wears a gilet over a green top and stands side on to the camera talking to an out of shot journalist with a serious expression. Behind him to the left is a large green tractor with yellow wheels. \n\n## 'Stressed and struggling' farmers call for fuel tax cuts\n\nFarmers say they need help to deal with rising prices while the Iran situation continues. [...] The US president says Nato \"wasn't there when we needed them\" during the Iran war.\n\nSir Keir Starmer speaks to members of the military. He is stood in the centre with military personnel in uniform to his left and right. Sir Keir is wearing a dark-coloured suit, a white shirt and a dark-coloured tie.\n\n## Starmer says much work remains to make US-Iran ceasefire hold\n\nThe prime minister says fully reopening the Strait of Hormuz would help \"stabilise\" prices in the UK.\n\nA woman walks a dog past a building in Tel Aviv, central Israel, that was heavily damaged by an Iranian ballistic missile strike, after the US and Iran agreed a two-week ceasefire (8 April 2026)\n\n## Israel backs US-Iran ceasefire but Netanyahu's war goals remain unfulfilled [...] Saeed Khatibzadeh pictured at a press conference in Tehran in 2022\n\n## US must choose 'between war and ceasefire', Iranian minister tells BBC\n\nIran's deputy foreign minister says Israeli strikes on Lebanon are a 'grave violation' of the deal.\n\nAli Capper stands on her farm, looking at the camera wearing a dark blue jacket over a red top. She has cropped brown hair.\n\n## UK farmers warn Iran ceasefire too late to stop higher food costs\n\nDespite a ceasefire, the Iran war will continue to affect farmers' costs and therefore food prices.\n\nForeign Secretary Yvette Cooper hosts a virtual meeting at the Foreign, Commonwealth and Development Office (FCDO) in Westminster. She wears black glasses, and a Union Jack is visible behind her.",

"score": 0.7171114

}

]

}

}

]

},

"logprobs": null,

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}

],

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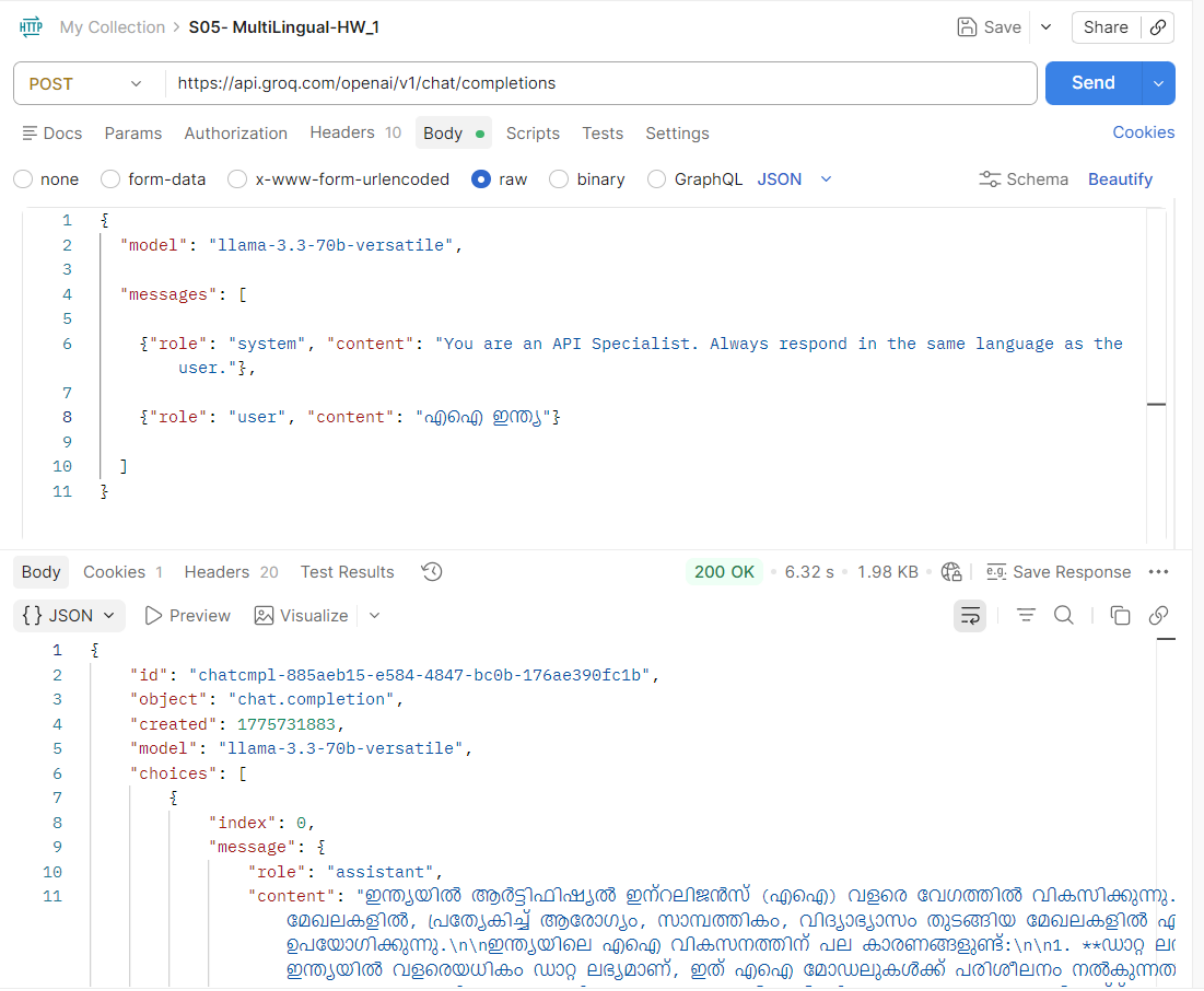
## 4. S05- MultiLingual-HW




**Request:**

```
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  "messages": [

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    {"role": "user", "content": "എഐ ഇന്ത്യ"}

  ]
}
```

**Response:**

{
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"created": 1775731883,
"model": "llama-3.3-70b-versatile",
"choices": [
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"role": "assistant",
"content": "ഇന്ത്യയിൽ ആർട്ടിഫിഷ്യൽ ഇന്റലിജൻസ് (എഐ) വളരെ വേഗത്തിൽ വികസിക്കുന്നു. വിവിധ മേഖലകളിൽ, പ്രത്യേകിച്ച് ആരോഗ്യം, സാമ്പത്തികം, വിദ്യാഭ്യാസം തുടങ്ങിയ മേഖലകളിൽ എഐ ഉപയോഗിക്കുന്നു.\n\nഇന്ത്യയിലെ എഐ വികസനത്തിന് പല കാരണങ്ങളുണ്ട്:\n\n1. **ഡാറ്റ ലഭ്യത**: ഇന്ത്യയിൽ വളരെയധികം ഡാറ്റ ലഭ്യമാണ്, ഇത് എഐ മോഡലുകൾക്ക് പരിശീലനം നൽകുന്നതിനും മെച്ചപ്പെടുത്തുന്നതിനും സഹായിക്കുന്നു.\n2. **സർക്കാർ പിന്തുണ**: ഇന്ത്യാ ഗവൺമെന്റ് എഐ ഗവേഷണത്തിനും വികസനത്തിനും പിന്തുണ നൽകുന്നു, ഇത് എഐ വികസനത്തിന് പ്രോത്സാഹനം നൽകുന്നു.\n3. **വ്യവസായ താൽപ്പര്യം**: ഇന്ത്യയിലെ പല വ്യവസായങ്ങളും എഐ സാങ്കേതികവിദ്യ ഉപയോഗിച്ച് തങ്ങളുടെ പ്രവർത്തനങ്ങൾ മെച്ചപ്പെടുത്താൻ ശ്രമിക്കുന്നു.\n\nഇന്ത്യയിലെ എഐ വികസനത്തിന്റെ ചില സവിശേഷതകൾ ഇതാ:\n\n* **ആരോഗ്യ സംരക്ഷണം**: ഇന്ത്യയിലെ ആരോഗ്യ സംരക്ഷണ മേഖലയിൽ എഐ വ്യാപകമായി ഉപയോഗിക്കുന്നു, രോഗനിർണ്ണയം, ചികിത്സ, രോഗപ്രതിരോധം തുടങ്ങിയ മേഖലകളിൽ.\n* **സാമ്പത്തികം**: ഇന്ത്യയിലെ സാമ്പത്തിക മേഖലയിൽ എഐ ഉപയോഗിക്കുന്നു, പണമടയ്ക്കൽ, സാമ്പത്തിക പ്രവചനം, റിസ്ക് മാനേജ്മെന്റ് തുടങ്ങിയ മേഖലകളിൽ.\n* **വിദ്യാഭ്യാസം**: ഇന്ത്യയിലെ വിദ്യാഭ്യാസ മേഖലയിൽ എഐ ഉപയോഗിക്കുന്നു, വിദ്യാർത്ഥികളുടെ പ്രകടനം മെച്ചപ്പെടുത്തുന്നതിനും അധ്യാപകരുടെ ജോലി ലഘൂകരിക്കുന്നതിനും.\n\nഇന്ത്യയിലെ എഐ വികസനത്തിന് പല വെല്ലുവിളികളുമുണ്ട്, പ്രത്യേകിച്ച് ഡാറ്റ ഗുണനിലവാരം, എഐ മോഡലുകളുടെ വിശ്വാസ്യത, എഐ സാങ്കേതികവിദ്യയുടെ ദുരുപയോഗം തുടങ്ങിയ കാര്യങ്ങൾ. എന്നിരുന്നാലും, ഇന്ത്യയിലെ എഐ വികസനം വളരെ വേഗത്തിൽ മുന്നേറുന്നു, ഇത് ഇന്ത്യയുടെ സാമ്പത്തികവും സാമൂഹികവും പരിവർത്തനത്തിന് സഹായിക്കും."
},

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